

EVENTFORCE MANAGEMENT SYSTEM

Phase 5 Comprehensive Documentation: Deployment, Documentation & Maintenance

Project Track: Salesforce CRM Enterprise Solution (Academic & Professional Specification)

Document Focus: Phase 5 – Deployment Strategy, System Maintenance, Documentation & Demo Video Guidelines

Target Environment: Developer Org (Simulated Production Migration)

Document Status: Final Technical Deliverable & Operations Manual

1. Executive Summary & Project Overview

The EventForce Management System is an end-to-end cloud CRM platform built on Salesforce to automate, optimize, and scale corporate and private event management operations. The platform addresses core industry challenges including vendor coordination delay, automated client billing, attendee registration tracking, resource allocation, and feedback gathering.

1.1 High-Level Business Objectives

- **Operational Efficiency (Centralized Event Management):** Unify event scheduling, venue logistics, client management, and vendor assignments into a single, cohesive Salesforce instance.
- **Process Automation (Automation & SLA Enforcement):** Utilize Salesforce Flow, Apex triggers, and Approval Processes to automate 3-day pre-event reminders, budget approvals, and status transitions.
- **Data Governance (Granular Security & Compliance):** Enforce role-based access control (RBAC) across Event Managers, Logistics Teams, and Executive Leadership using Profiles, Roles, and Sharing Rules.
- **Analytics & Insights (Actionable Business Intelligence):** Deliver real-time executive dashboards and performance reports analyzing revenue, vendor performance, and customer satisfaction scores.

2. Summary of Implementation Phases (Phases 1–4)

Prior to Phase 5, the EventForce Management System underwent rigorous requirement gathering, data modeling, backend architecture, custom user interface design, and security engineering.

Phase	Core Focus Area	Key Deliverables & Artifacts
Phase 1	Requirement Analysis & Data Model	Detailed Business Requirement Document (BRD), ERD diagram, Custom Objects (Event__c, Client__c, Vendor__c, Event_Feedback__c), custom fields, picklists, and master-detail/lookup relationships.
Phase 2	Backend Logic & Automations	Salesforce Record-Triggered Flows, 3-Day Event Reminder Scheduled Flow, Validation Rules (e.g., Budget > 0, Event Date in future), Apex Triggers for budget roll-up, and Automated Approval Processes.
Phase 3	UI Customization & Page Layouts	Custom Lightning App ('EventForce Studio'), optimized Page Layouts, Lightning Record Pages with Dynamic Action/Visibility rules, Compact Layouts, and List Views.
Phase 4	Security & Quality Assurance	Role Hierarchy (CEO -> Regional Manager -> Event Exec), Custom Profiles, Permission Sets, Sharing Rules, Test Classes with >85% code coverage, and UAT execution.

3. Phase 5: Deployment Strategy & DevOps Simulation

In the scope of the EventForce Management System project, all configurations, customizations, and automations were designed, implemented, and validated inside a Salesforce Developer Edition Org. This section outlines the enterprise deployment methodology simulated for production readiness.

3.1 Environment Rationale: Developer Org vs. Enterprise Pipeline

Actual production deployment (migration from development to a live environment) was not directly executed in a live production org due to standard Salesforce architectural constraints:

- **Standalone Environment Architecture:** Developer Edition orgs are standalone, isolated environments that lack direct outbound change-set connections to live enterprise production orgs.
- **Enterprise CI/CD Requirements:** Enterprise implementations mandate a multi-tiered sandbox pipeline (Developer Sandbox -> Partial Copy Sandbox -> Full Copy Sandbox -> Production) governed by continuous integration tools

3.2 Enterprise Migration & Metadata Management Strategy

To prepare EventForce for commercial migration, all platform components are structured into metadata packages. The deployment workflow adheres to standard enterprise categorizations:

Metadata Category	Salesforce Metadata API Types	EventForce Artifacts Included
Data Architecture	CustomObject, CustomField, RecordType	Event__c, Client__c, Vendor__c, Feedback__c, relationship fields.
Business Logic	ApexClass, ApexTrigger, Flow	EventBudgetTrigger, ThreeDayReminderFlow, EventApprovalProcess.
User Interface	CustomApplication, FlexiPage, Layout	EventForce App, Event_Record_Page, Vendor_Layout.
Security & Access	Profile, PermissionSet, CustomRole	Event_Manager_Profile, Vendor_Read_Permission, Executive_Role.
Analytics	Report, Dashboard	Event_Revenue_Report, Executive_KPI_Dashboard.

3.3 Deployment Tooling & CI/CD Pipeline

For enterprise migration, EventForce utilizes a modern Salesforce DX (SFDX) and GitHub Actions pipeline:

- **Source Code Versioning:** All metadata is extracted using SFDX CLI (sf project retrieve start) and committed to a Git repository.
- **Automated Testing & Validation:** Continuous Integration (CI) triggers automated unit testing (sf apex run test) whenever code is merged to the integration branch.
- **Change Sets & DevOps Center:** For declarative deployments, Salesforce Change Sets or DevOps Center track dependent metadata to prevent deployment failures due to missing components.

4. Maintenance, Monitoring & Troubleshooting

Post-deployment operational stability requires continuous platform monitoring, proactive data hygiene, and structured troubleshooting protocols to guarantee high system availability and user adoption.

4.1 Ongoing Maintenance Activities

- **Data Quality & Hygiene:** Weekly deduplication of Client and Vendor records, along with periodic archival of completed historical events.
- **User Offboarding & Access Audits:** Quarterly audits of active user licenses, profile assignments, and permission set allocations.
- **Salesforce Release Readiness:** Thrice-yearly review of Salesforce major releases (Spring, Summer, Winter) to evaluate new feature capabilities and test backward compatibility.

4.2 System Monitoring Framework

- **Apex Exception Logs:** Monitoring Developer Console and Setup debug logs for runtime uncaught Apex exceptions.
- **Flow Engine Monitoring:** Reviewing 'Paused and Failed Flow Executions' in Setup to catch unhandled fault paths or null pointer errors.
- **Scheduled & Batch Job Tracking:** Auditing Apex Jobs (AsyncApexJob) to ensure the 3-day event reminder scheduled flow executes without queue limits.

4.3 Troubleshooting & Incident Resolution Guide

Issue Category	Common Root Cause	Diagnostic Tool / Area	Resolution & Corrective Action
User Access Denied	Missing Permission Set or OWD restriction	User Detail -> Permission Set Assignments	Grant required Permission Set or adjust Sharing Rules.
Validation Rule Block	Input data fails criteria (e.g., past date)	Validation Rule Setup	Adjust input format or refine Validation Rule error condition.
Flow Execution Fault	Null value passed into required field	Paused & Failed Flow Executions	Add Fault Path in Flow to handle null checks gracefully.
Apex CPU Time Limit	Unoptimized SOQL queries inside loops	Apex Debug Logs / Developer Console	Bulkify Apex Trigger code and optimize SOQL queries.

5. Project Documentation & Feature Evidence Specification

Comprehensive documentation guarantees knowledge transfer and long-term maintainability. Below is the blueprint of required system artifacts and screenshot evidence mandatory for final submission.

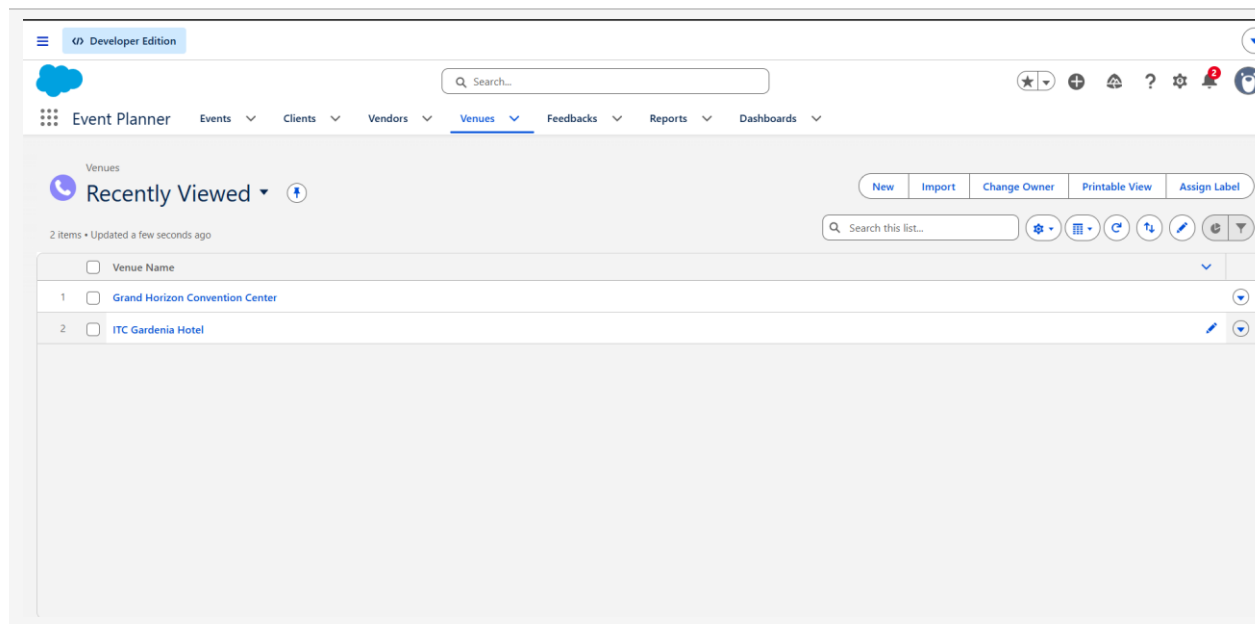
5.1 Core Artifact Inventory

- **Data Architecture:** Business Requirements & Data Model (ER Diagram mapping Event__c, Client__c, Vendor__c, Feedback__c).
- **Automation Architecture:** Automation Map (Flow Diagrams, Apex Trigger logic, Approval Process entry criteria).
- **Security Governance:** Security Matrix (Profile vs. Permission Set matrix and OWD configuration).

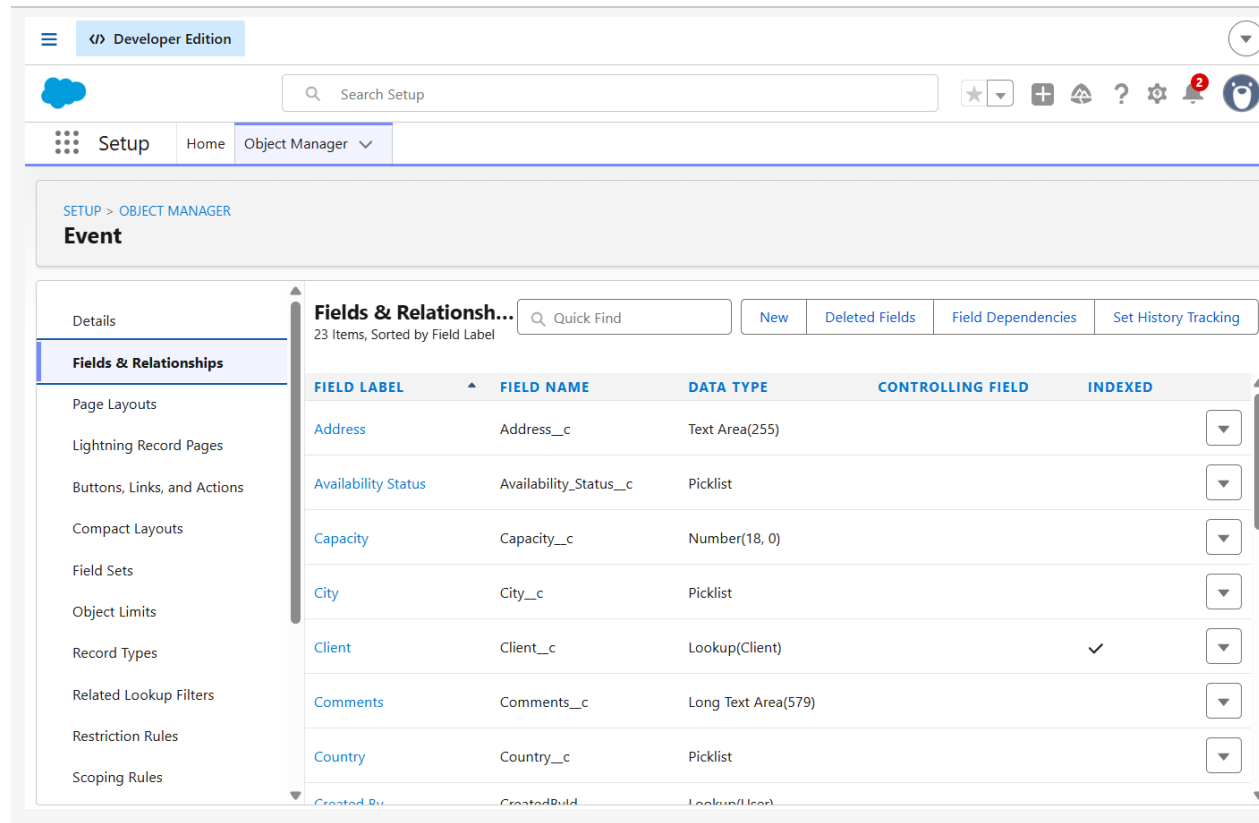
5.2 System Visual Evidence

As per project guidelines, the following visual verification screenshots are embedded into the official project deliverable:

[Screenshot 5.1: EventPlanner Custom App Launcher & App Layout] – Capturing custom navigation, primary tabs, and branded header.



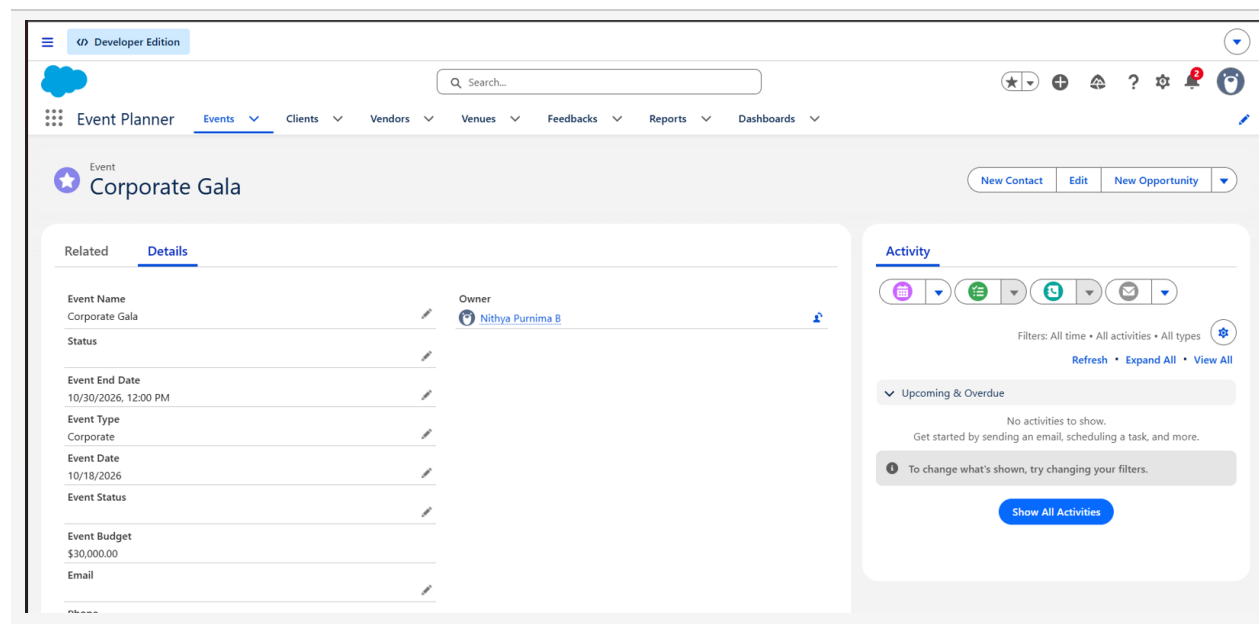
[Screenshot 5.2: Event__c & Vendor__c Object Manager Schema] – Displaying custom field types, picklist values, and lookup relationships.



This screenshot shows the Salesforce Object Manager interface for the 'Event' object. The left sidebar contains a navigation menu with options like Details, Fields & Relationships (selected), Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Restriction Rules, and Scoping Rules. The main area displays the 'Fields & Relationships' section, showing 23 items sorted by Field Label. A table lists the fields with columns for Field Label, Field Name, Data Type, Controlling Field, and Indexed status.

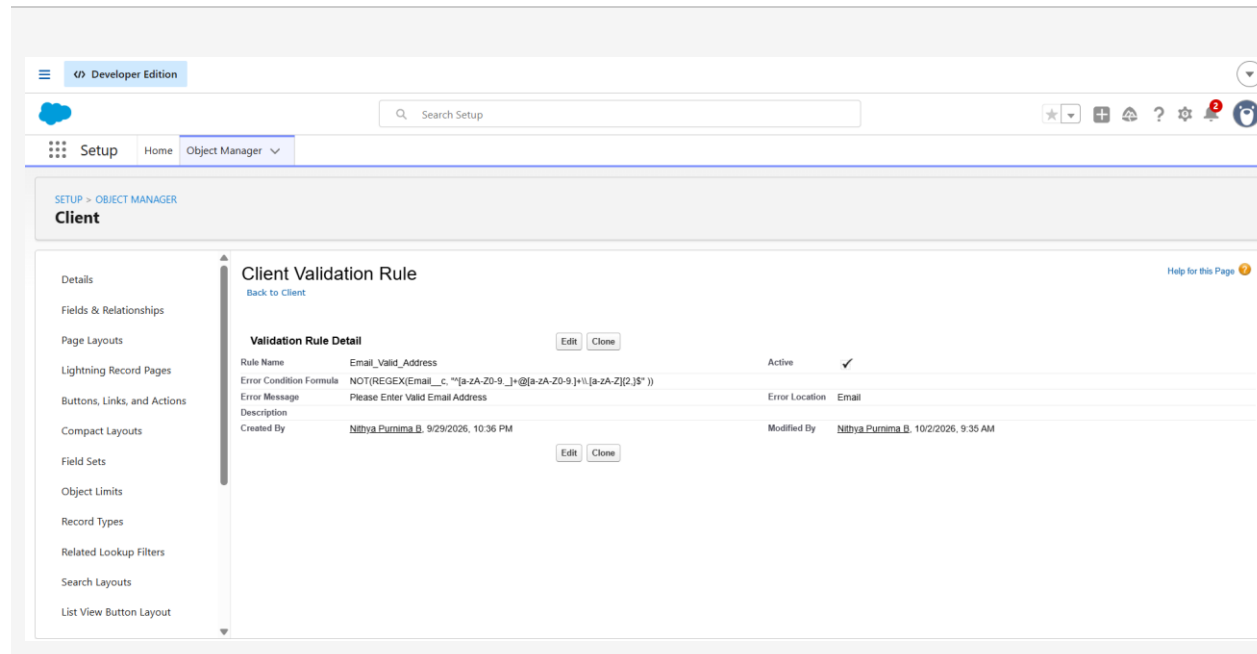
FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Address	Address__c	Text Area(255)		
Availability Status	Availability_Status__c	Picklist		
Capacity	Capacity__c	Number(18, 0)		
City	City__c	Picklist		
Client	Client__c	Lookup(Client)		✓
Comments	Comments__c	Long Text Area(579)		
Country	Country__c	Picklist		
Created By	CreatedBy__c	Lookup(User)		

[Screenshot 5.3: Event Lightning Record Page & Dynamic Layout] – Demonstrating dynamic fields, record highlights, and component visibility.

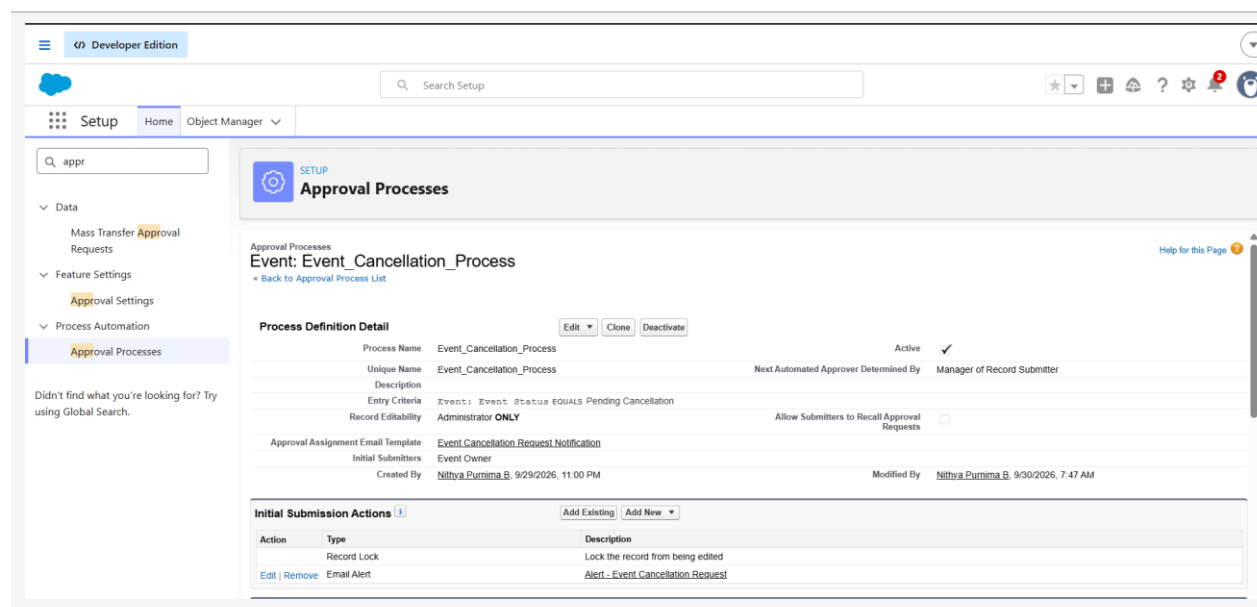


This screenshot shows the Salesforce Lightning Record Page for the 'Corporate Gala' event. The page is divided into two main sections: 'Details' and 'Activity'. The 'Details' section displays various fields for the event, including Event Name, Status, Event End Date, Event Type, Event Date, Event Status, Event Budget, and Email. The 'Activity' section shows a list of activities related to the event, with filters for 'All time', 'All activities', and 'All types'. A 'Show All Activities' button is visible at the bottom of the activity section.

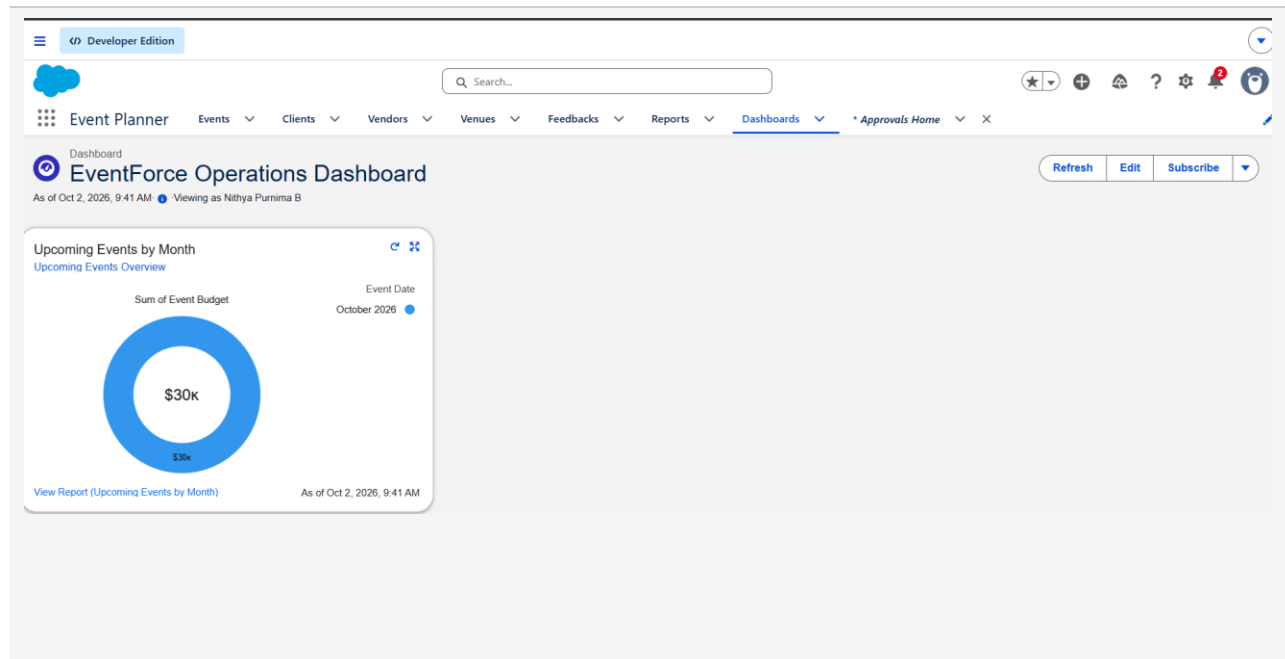
[Screenshot 5.4: Validation Rule Trigger in Action] – Showing error messaging when invalid budget or date criteria is entered.



[Screenshot 5.5: Event Approval Process Workflows] – Displaying submission, manager approval steps, and status field updates.



[Screenshot 5.6: Executive Dashboard & Analytics Reports] – Showcasing real-time corporate event analytics, revenue gauges, and charts.



6. Guidelines & Structural Script for Project Demo Video

The demo video provides a walkthrough of the EventForce Management System for stakeholders. Presenters must strictly follow the structured 8-step script and pro tips outlined below.

6.1 Step-by-Step Presentation Script (Target Duration: 5–10 Minutes)

Step	Section Title	Target Time	Demonstration Script & Key Visual Focus
1	Introduction	0:00 - 0:45	Introduce yourself, project title ('EventForce Management System'), and present high-level business goals.
2	App Overview	0:45 - 1:30	Navigate App Launcher, open EventForce Studio, show core navigation items (Events, Clients, Vendors).
3	UI Demonstration	1:30 - 3:00	Create a new Event record, demonstrate validation rules in action (trigger deliberate error), highlight dynamic layout.
4	Automation Showcase	3:00 - 5:00	Walk through the 3-Day Reminder Flow, trigger the Approval Process for high-budget events, and review Apex logic.
5	Security Setup	5:00 - 6:30	Explain Security Architecture: demonstrate Profiles, Permission Sets, Roles, and Sharing Settings.
6	Reports & Dashboards	6:30 - 7:45	Display Executive KPI Dashboard, drill down into Event Revenue and Vendor Satisfaction reports.
7	Debugging & Fixes	7:45 - 8:45	Discuss real development challenges encountered, how debug logs were used, and resolution of flow/validation errors.
8	Conclusion	8:45 - 9:30	Summarize core outcomes, key Salesforce learnings, and thank evaluators.

6.2 Presentation Quality & Technical Pro Tips

- **Voice & Audio Quality:** Use clear audio input, ensure low background noise, and speak in a clear, professional tone.
- **Display Resolution:** Record at 1080p HD resolution with clean browser zoom (100%-110%) for readability.

- **Logical Flow:** Strictly follow the logical sequence: Intro -> UI -> Automation -> Security -> Reports -> Conclusion.
- **Highlight Key Automations:** Dedicate extra focus to high-value automations such as the 3-Day Reminder Flow and multi-step Approval Process.

7. Conclusion & Project Outcomes

The EventForce Management System demonstrates how Salesforce can streamline and automate complex event-planning operations through customized objects, intelligent automations, robust security controls, and real-time analytics.

Project Achievements: Key Outcomes & Technical Learnings

- **Architectural Mastery:** Mastered enterprise relational data modeling in Salesforce using Master-Detail and Lookup relationships.
- **Automation Expertise:** Built complex declarative automations (Record-Triggered & Scheduled Flows) alongside programmatic Apex extensions.
- **Governance & Security:** Implemented enterprise security standards using Role Hierarchies, Profiles, and Organization-Wide Defaults.
- **Lifecycle Management:** Gained deep insight into full lifecycle CRM development—from BRD creation to production migration simulation.

Although constructed within a Salesforce Developer Edition Org for academic and demonstration purposes, this project accurately reflects real-world CRM implementation best practices. The system is fully tested, thoroughly documented, and prepared for future enterprise-level scaling and deployment.